

Predictors of COVID-19 Protective Behavior Before and After Mask Mandates: A Cross-National Analysis

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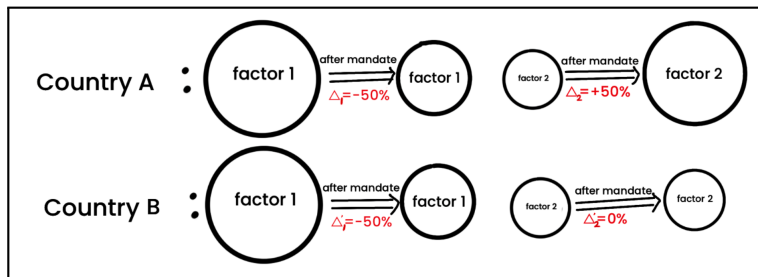
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1. Project background

- During **COVID-19**, protective behaviours such as **mask wearing**, social distancing, hand hygiene, and self-isolation were important for reducing transmission.
- Compliance was shaped not only by policy rules, but also by risk perception, trust, demographic characteristics, wellbeing, and broader social context.
- These predictors may operate **differently across countries** because policy environments, institutions, culture, and public expectations vary.

2. Research questions

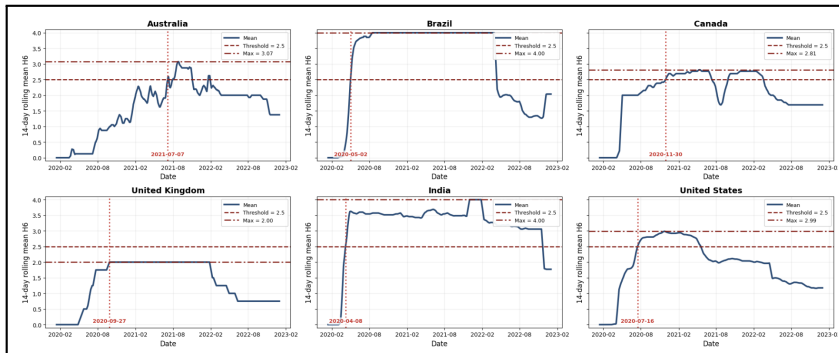
- Within each country, do the key predictors of compliance with COVID-19 protective behaviours change before and after mask mandates?
- Are these changes consistent across countries, or do they vary?
- Is there a set of common cross-national predictors, alongside country-specific predictors?



1. Behaviour and policy datasets

- **YouGov dataset:** repeated survey responses about people's COVID-19 behaviours, attitudes, and background characteristics.
 - Behaviour: whether people wore masks, kept physical distance, washed hands...
 - Predictor: age group, gender, confidence in health services...
- **OxCGRT dataset:** This dataset contains the strictness with which the government has enforced its mask-wearing policy.
- The two sources are linked by **country, region and survey date**.

2. Policy timing figure



This figure show how mask policy intensity changed over time, and how the mandate threshold was used to define the before and after mandate periods.

3. Country coverage

The current cleaned cross-national sample includes six countries:

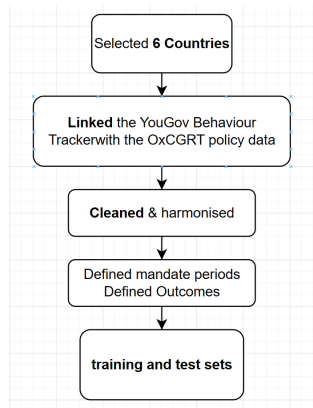
Country code	Country name
AUS	Australia
BRA	Brazil
CAN	Canada
CHN	China
GBR	United Kingdom
IND	India
USA	United States

China was excluded from the current analysis because several key variables needed for the harmonised workflow were not available.

What Have I Done So Far?

1. Data pre-processing

- For the YouGov and OxCGRT files, various formatting issues have been resolved.
- Mapped survey regions to policy regions: Convert all place names in both datasets to lower-case English letters, then match them one-to-one.
- Defined before/after mandate periods.
- Defined two outcomes: Face-mask-wearing and Overall-protective-behaviour.
- Cleaned the data: Removing columns and samples with too many missing values, encoding the data, making column names more readable, and splits.



2. About Model

Model 1: Random Forest

A robust tree-based ensemble model; selected for its stable performance on mixed survey data, resistance to noise, and ability to rank feature importance.

Model 2: XGBoost

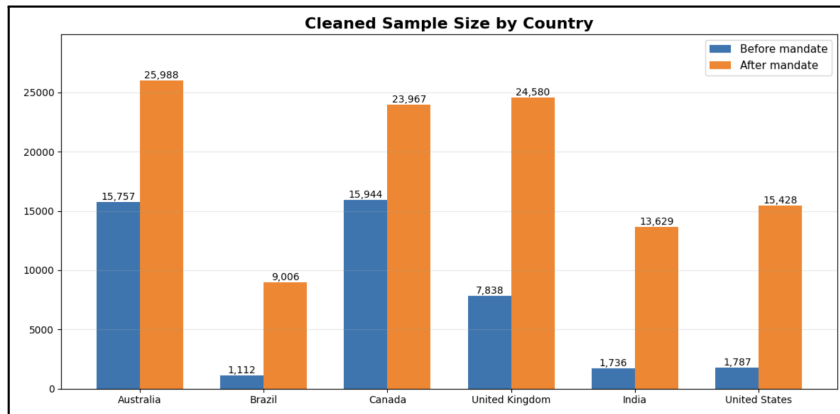
A gradient-boosted tree model; selected for its strong predictive accuracy, ability to capture nonlinear patterns, and built-in regularization.

Metric: AUC

AUC means Area Under the ROC Curve; selected because it can measure model separation across all classification thresholds. AUC ranges from 0 to 1. A higher AUC indicates better separation between compliant and non-compliant respondents.

1. Cleaned sample size

The preprocessing pipeline produced cleaned before- and after-mandate datasets for all six countries.



2. Train-test split

The next step was to prepare data for modelling.

- I split the data separately for each country, policy period, and outcome.
- I used an 80/20 train-test split.
- Stratification was used where possible to preserve the class balance of the outcome.
- Stratification was based on the outcome class, so the train and test sets kept similar proportions of compliant and non-compliant respondents.

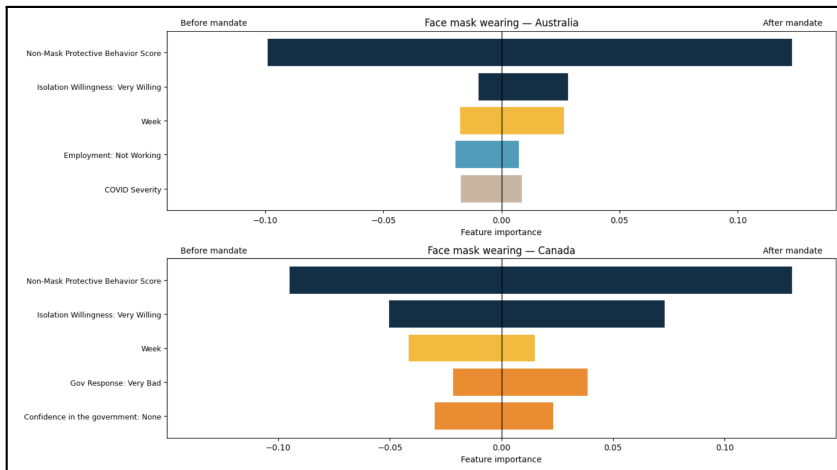
Dimension	Values	Number
Countries	AUS, BRA, CAN, GBR, IND, USA	6
Policy periods	Before mandate, after mandate	2
Outcomes	Face mask, overall protective behaviour	2
Total modelling tasks	$6 \times 2 \times 2$	24

3. Model Performance (AUC)

Country	Outcome	RF	XGboost	Country	Outcome	RF	XGboost
Australia	Face mask wearing Before mandate	0.844	0.854	UK	Face mask wearing Before mandate	0.755	0.760
	Face mask wearing After mandate	0.909	0.912		Face mask wearing After mandate	0.695	0.705
	Protective behaviour Before mandate	0.767	0.772		Protective behaviour Before mandate	0.796	0.797
	Protective behaviour After mandate	0.826	0.831		Protective behaviour After mandate	0.846	0.857
Brazil	Face mask wearing Before mandate	0.783	0.785	India	Face mask wearing Before mandate	0.757	0.764
	Face mask wearing After mandate	0.652	0.653		Face mask wearing After mandate	0.856	0.851
	Protective behaviour Before mandate	0.713	0.705		Protective behaviour Before mandate	0.823	0.801
	Protective behaviour After mandate	0.829	0.827		Protective behaviour After mandate	0.827	0.831
Canada	Face mask wearing Before mandate	0.854	0.854	US	Face mask wearing Before mandate	0.846	0.853
	Face mask wearing After mandate	0.813	0.816		Face mask wearing After mandate	0.855	0.862
	Protective behaviour Before mandate	0.836	0.840		Protective behaviour Before mandate	0.797	0.802
	Protective behaviour After mandate	0.833	0.834		Protective behaviour After mandate	0.857	0.862

4. Feature importance

Feature importance indicates how much each predictor contributes to the model's prediction. A higher importance score means that the factor plays a stronger role (Calculated after model training based on the contribution of each variable to the predicted results).



Summary of current progress

- YouGov behavioural data and OxCGRT mask-policy data have been cleaned, linked, and split for six analysable countries.
- 24 modelling tasks have been prepared across countries, outcomes, and mandate periods.
- Random Forest and XGBoost results provide the basis for comparing shared and country-specific predictors.

Next step

- Analyse the multi-country results in detail and begin writing the final report.

All data and code

All code can be found in my GitHub: <https://github.com/ZixaunZhao/COVID-19-Protective-Behavior-Cross-National-Analysis.git>